

Algebra Homework 1

Problem 1. Let $K, L \subseteq E$ be subfields of a field E , and put

$$R = \left\{ \sum_{j=1}^t k_j \ell_j : t \geq 1, k_j \in K, \ell_j \in L \right\}.$$

Write KL for the compositum of K and L in E .

- (i) Prove that $KL = R$ if every element of K is algebraic over L .
- (ii) Give an example in which $KL \neq R$.

Proof. (i) The set R is a subring of E containing both K and L , and $R \subseteq KL$. It therefore suffices to prove that R is a field. Let $0 \neq a \in R$, and write $a = \sum_{i=1}^t k_i \ell_i$. Since each k_i is algebraic over L , then $L(k_1, \dots, k_t)/L$ is algebraic. As a belongs to this algebraic extension, write its monic minimal polynomial as

$$f(X) = \sum_{j=0}^d c_j X^j \in L[X], \quad \text{with } c_d = 1.$$

Since $a \neq 0$, we have $c_0 \neq 0$. Dividing $f(a) = 0$ by a and rearranging gives

$$a^{-1} = -c_0^{-1} \sum_{j=1}^d c_j a^{j-1} \in R.$$

Hence R is a field containing K and L , so $KL \subseteq R$. Together with $R \subseteq KL$, this proves $KL = R$.

(ii) Let x, y be algebraically independent over $\mathbb{Q}$, and let

$$K = \mathbb{Q}(x), \quad L = \mathbb{Q}(y).$$

We first prove that $KL = \mathbb{Q}(x, y)$. Since E contains K and L , the definition of the compositum gives $KL \subseteq E$. Conversely, KL contains $\mathbb{Q}$, x , and y , so it contains every polynomial in x, y with rational coefficients. As a field, it also contains the inverse of every nonzero such polynomial. It therefore contains $\mathbb{Q}(x, y)$, which proves the equality.

On the other hand,

$$R = \left\{ \frac{P(x, y)}{A(x)B(y)} : P \in \mathbb{Q}[x, y], 0 \neq A \in \mathbb{Q}[x], 0 \neq B \in \mathbb{Q}[y] \right\}.$$

Suppose that $(x + y)^{-1} \in R$. The preceding description yields the identity

$$A(x)B(y) = (x + y)P(x, y).$$

Substituting $y = -x$ yields $A(x)B(-x) = 0$ in $\mathbb{Q}[x]$. Both factors are nonzero, contradicting the fact that $\mathbb{Q}[x]$ is an integral domain. Thus $(x + y)^{-1} \notin R$, and $R \neq KL$.

In this example, R is actually naturally isomorphic to $\mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{Q}(y)$. Indeed, multiplication in E induces a $\mathbb{Q}$ -algebra homomorphism

$$\mu : \mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{Q}(y) \longrightarrow \mathbb{Q}(x, y), \quad k \otimes \ell \longmapsto k\ell,$$

whose image is R . To prove injectivity, let $z \in \ker \mu$. After choosing common denominators in the two factors, write

$$z = \sum_i \frac{p_i(x)}{A(x)} \otimes \frac{q_i(y)}{B(y)},$$

where $p_i \in \mathbb{Q}[x]$, $q_i \in \mathbb{Q}[y]$, and A, B are nonzero. Expanding the numerators and using bilinearity, we write z as

$$z = \sum_{r,s} c_{rs} \frac{x^r}{A(x)} \otimes \frac{y^s}{B(y)}, \quad c_{rs} \in \mathbb{Q},$$

with only finitely many nonzero coefficients. Then

$$0 = \mu(z) = \frac{\sum_{r,s} c_{rs} x^r y^s}{A(x)B(y)}.$$

Algebraic independence of x, y implies that every c_{rs} is zero. Hence $z = 0$, and μ induces the isomorphism

$$R \cong \mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{Q}(y).$$

Problem 2. Let $\alpha = \sqrt[3]{2} \in \mathbb{R}$ and let $\xi \in \mathbb{C}$ be a primitive ninth root of unity. Compute $\text{Gal}(\mathbb{Q}(\alpha, \xi)/\mathbb{Q})$.

Proof. Let $E = \mathbb{Q}(\alpha, \xi)$, $L = \mathbb{Q}(\xi)$. By rational root theorem, the polynomial $X^3 - 2$ is irreducible over $\mathbb{Q}$. Moreover, its complex roots are $\alpha, \alpha\omega$ and $\alpha\omega^2$, where ω is the cubic root of unity. Since $\alpha\omega$ and $\alpha\omega^2$ do not belong to $\mathbb{Q}(\alpha)$, thus $\mathbb{Q}(\alpha)/\mathbb{Q}$ is not normal and hence not Galois.

The element ξ is a root of $g(X) = X^6 + X^3 + 1$. Also by rational root theorem, $g(X)$ is irreducible over $\mathbb{Q}$. It follows that $[L : \mathbb{Q}] = 6$. Since L is the splitting field of $X^9 - 1$, it is Galois over $\mathbb{Q}$. As shown in Problem 3, the six maps $\xi \mapsto \xi^u$, where $u \in (\mathbb{Z}/9\mathbb{Z})^\times$, are its automorphisms. Therefore

$$\text{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/9\mathbb{Z})^\times \cong C_6.$$

Every subgroup of an abelian group is normal. By the fundamental theorem of Galois theory, every intermediate field of $L/\mathbb{Q}$ is therefore Galois over $\mathbb{Q}$. Since $\mathbb{Q}(\alpha)/\mathbb{Q}$ is not Galois, we have $\alpha \notin L$. If $X^3 - 2$ were reducible over L , it would have a root in L , since its degree is three. This root would be $\alpha\omega^j$ for some $j \in \{0, 1, 2\}$. As $\omega = \xi^3 \in L$, this would imply $\alpha \in L$. Hence $X^3 - 2$ is irreducible over L , and it follows that

$$[E : \mathbb{Q}] = [E : L][L : \mathbb{Q}] = 3 \cdot 6 = 18.$$

Then the field E is the splitting field of $(X^3 - 2)(X^9 - 1)$ over $\mathbb{Q}$. Over a characteristic zero field,

this automatically implies that $E/\mathbb{Q}$ is Galois.

Since $X^3 - 2$ is irreducible over L and splits in E , there is an L -automorphism σ of E satisfying

$$\sigma(\alpha) = \alpha\omega, \quad \sigma(\xi) = \xi.$$

To construct a second automorphism, use the identification $E \cong L[T]/(T^3 - 2)$. Apply $\xi \mapsto \xi^2$ to the coefficients and leave the class of T fixed. Since $T^3 - 2$ has rational coefficients, this coefficient automorphism preserves the ideal $(T^3 - 2)$ and induces an automorphism τ of E with

$$\tau(\alpha) = \alpha, \quad \tau(\xi) = \xi^2.$$

The orders of σ and τ are 3 and 6, respectively. The 18 automorphisms $\sigma^r \tau^s$, with $0 \leq r < 3$ and $0 \leq s < 6$, are distinct. Since $[E : \mathbb{Q}] = 18$, these are all the automorphisms of $E/\mathbb{Q}$. Notice that $\tau\sigma\tau^{-1} = \sigma^{-1}$, so

$$\text{Gal}(E/\mathbb{Q}) \cong C_3 \rtimes C_6,$$

Moreover, we consider about $H = \langle \sigma, \tau^3 \rangle$ and $J = \langle \tau^2 \rangle$. The group H has order 6 and is isomorphic to S_3 , since τ^3 has order 2 and conjugates σ to σ^{-1} . The group J is cyclic of order 3. It commutes with H because conjugation by τ^2 fixes σ . Easy check shows that $H \cap J = \{1\}$. Finally, H and J generate the whole group, since $\tau = \tau^3(\tau^2)^{-1}$. Therefore

$$\text{Gal}(E/\mathbb{Q}) \cong S_3 \times C_3.$$

Problem 3. Let $n \geq 1$ and let ζ_n be a primitive n th root of unity. Prove that $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$.

Proof. Let f be the monic minimal polynomial of ζ_n over $\mathbb{Q}$. By Gauss's lemma, one has

$$X^n - 1 = f(X)g(X), \quad f, g \in \mathbb{Z}[X].$$

Every root of f is a primitive n th root of unity, since field embeddings preserve multiplicative order.

We show that, for every root η of f and every prime $p \nmid n$, the number η^p is also a root of f . Suppose otherwise. Since $(\eta^p)^n = 1$, we have $g(\eta^p) = 0$, and therefore $f \mid g(X^p)$. Division by the monic polynomial f gives

$$g(X^p) = f(X)h(X), \quad h \in \mathbb{Z}[X].$$

Reducing modulo p , we obtain

$$\bar{g}(X)^p = \bar{f}(X)\bar{h}(X).$$

Since $\bar{f}$ is nonconstant, this identity implies that $\bar{f}$ and $\bar{g}$ have a common irreducible factor. Thus

$$X^n - 1 = \bar{f}(X)\bar{g}(X)$$

has a repeated factor in $\mathbb{F}_p[X]$. This contradicts

$$\gcd(X^n - 1, nX^{n-1}) = 1,$$

which holds because $p \nmid n$.

For every positive integer k coprime to n , successive application of this result to the prime factors of k shows that $f(\zeta_n^k) = 0$. Hence the roots of f are precisely the primitive n th roots of unity. Since $f \mid X^n - 1$ has no repeated roots in characteristic zero, we conclude that

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \deg f = \varphi(n).$$

Problem 4. Let F be a field.

- (i) Prove that there exists an algebraic extension E/F in which every nonconstant polynomial in $F[X]$ has a root.
- (ii) Prove that there exists an algebraic extension $\overline{F}/F$ such that $\overline{F}$ is algebraically closed.

Proof. (i) Let $\mathcal{P}$ be the set of monic nonconstant polynomials in $F[X]$. For each $f \in \mathcal{P}$, introduce an indeterminate X_f , and define

$$A = F[X_f : f \in \mathcal{P}], \quad I = (f(X_f) : f \in \mathcal{P}) \subseteq A.$$

Suppose that $I = A$. Then there would be distinct $f_1, \dots, f_m \in \mathcal{P}$ and $g_1, \dots, g_m \in A$ such that

$$1 = \sum_{j=1}^m g_j f_j(X_{f_j}).$$

Set $K_0 = F$. For $1 \leq j \leq m$, choose a monic irreducible factor q_j of f_j in $K_{j-1}[T]$, and put $K_j = K_{j-1}[T]/(q_j(T))$. Each K_j/K_{j-1} is finite, and the class of T is a root of f_j . Identifying each field with its image in the next, $K = K_m$ is a finite extension of F containing a root a_j of every f_j .

Define an F -algebra homomorphism $\psi : A \rightarrow K$ by sending X_{f_j} to a_j for $1 \leq j \leq m$ and sending every other indeterminate to zero. Since $f_j(a_j) = 0$ for every j , applying ψ to the identity for 1 gives $1 = 0$, a contradiction. Therefore I is proper.

By **Zorn's lemma**, I is contained in a maximal ideal M of A . Set $E = A/M$. Every nonzero element of F is a unit in A , so $M \cap F = \{0\}$. The quotient map therefore embeds F into the field E . For each $f \in \mathcal{P}$, let $a_f = X_f + M$. Since $f(X_f) \in M$, we have $f(a_f) = 0$. Thus every a_f is algebraic over F . Every element of E is a polynomial expression in finitely many of the a_f , and hence lies in a finite extension $F(a_{f_1}, \dots, a_{f_t})$. It follows that E/F is algebraic. Every monic nonconstant polynomial over F has a root in E . Dividing a nonconstant polynomial by its leading coefficient proves the assertion for all nonconstant polynomials in $F[X]$.

- (ii) Set $E_0 = F$. Applying (i) successively, choose algebraic extensions E_{r+1}/E_r in which every

nonconstant polynomial over E_r has a root. Identify each field with its image in the next, and put

$$\overline{F} = \bigcup_{r \geq 0} E_r.$$

Any finite subset of the union lies in some E_r , so $\overline{F}$ is a field. Each E_r/F is algebraic by transitivity, and every element of $\overline{F}$ belongs to some E_r . Hence $\overline{F}/F$ is algebraic.

Let $P \in \overline{F}[X]$ be nonconstant. Its finitely many coefficients lie in a common E_r , so $P \in E_r[X]$. By construction, P has a root in $E_{r+1} \subseteq \overline{F}$. Thus every nonconstant polynomial over $\overline{F}$ has a root in $\overline{F}$. Hence $\overline{F}$ is algebraically closed.